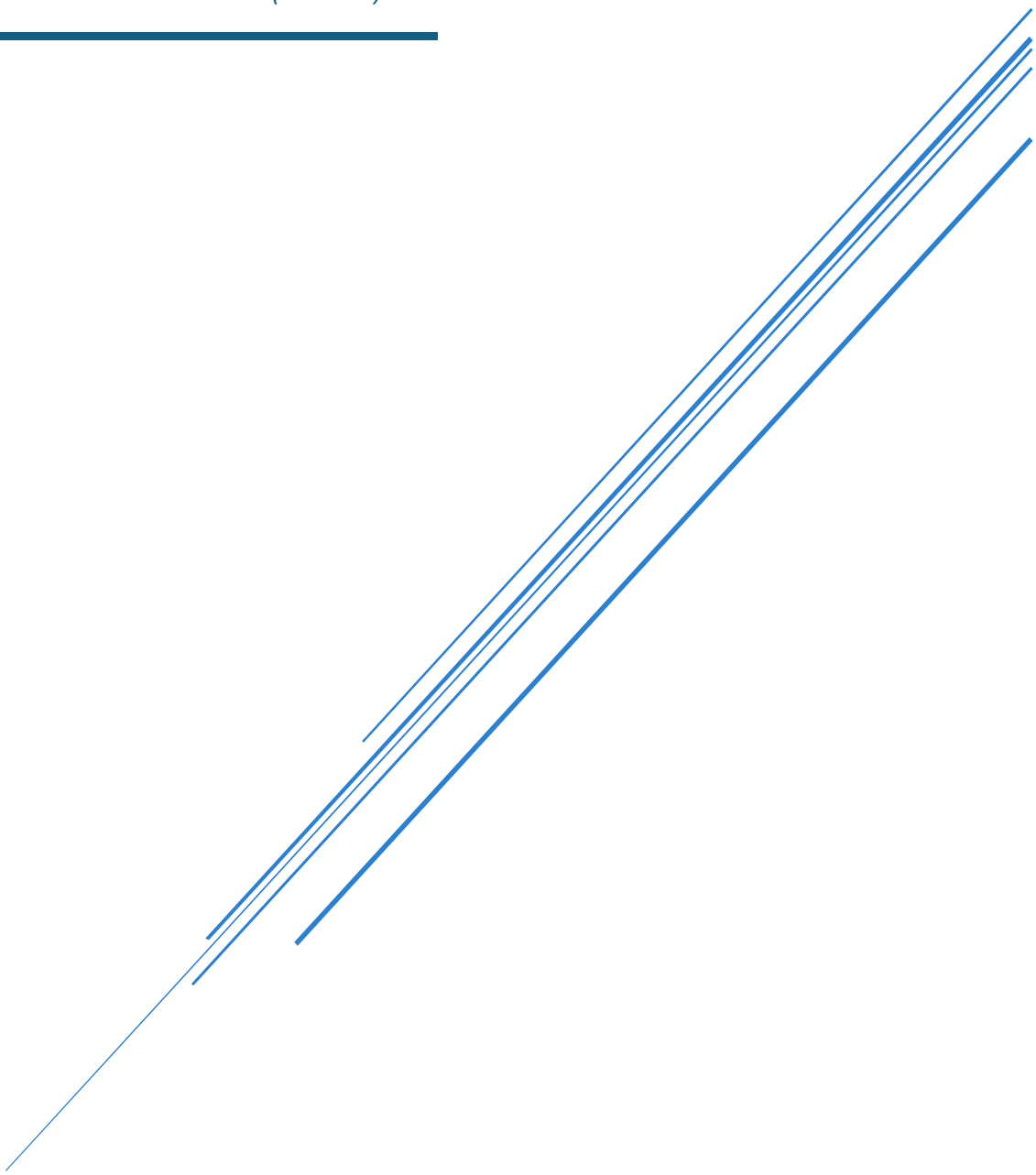
COA GROUP A

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Date of Submission –

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Section 1:

Introduction:

This document contains our answers to the questions of the Computer Architecture project, specifically:

- What is the CPU Machine Cycle?
- What are the fundamental concepts of RAM?
- What are the differences in the architecture of Intel, SnapDragon and Mac Silicon and their CPU's and what separates them from each other?
- What makes HDD, Tape Drives and SSD different from each other and how their inner workings make them excellent in in situations where they are most desired in the modern era?

What is the CPU Machine Cycle:

The Central Processing Unit is he primary component of a computer that is used to process data and instructions.

A CPU cycle refers to one tick of the processors internal clock, during which the CPU can execute either a part of or the entirety of a instruction in a single cycle depending on the architecture.

The CPU machine cycle is the sequence of steps that the CPU performs to execute instructions given to it. The 4 typical steps are: Fetch, Decode, Execute and Store.

Fetch: The CPU retrieves the next instruction from memory as data and temporarily stores it in the registers of the CPU.

Decode: The instruction data is translated into a set of electrical signals that the components of the CPU understand and recognise as a specific action to perform.

Execute: The CPU carries out the requested instruction, this can include performing calculations or retrieving a file from storage or accessing data.

Store: Once the instruction has been executed, the result of the operation is then stored back into memory, after this the cycle starts again with the CPU fetching the request that was next in the queue.

Section 2

What is RAM:

RAM which stands for Random Access Memory, is the computers short term memory. It is volatile memory which just means that if your computer device loses power, all data stored in the RAM is lost.

The SSD (Solid State Drive) and HDD (Hard Disk Drive) of a computer are like a massive warehouse full of boxes, RAM is the countertop right next to the worker (the CPU).

The purpose of the RAM: A worker can't do their job if they must run to the warehouse every time they need a single tool. They bring the tools they need now onto the countertop.

The RAM: is the countertop where the worker keeps the tools they are using**.**

If the countertop is too small, the worker keeps walking back to the warehouse, which is very time consuming.
If the desk is large, everything is right in front of them which means work can be done faster.

So that is why RAM is so important, because it allows the computer to work quickly.

How it works:

At a microscopic level, the RAM chip is like a massive grid of millions of cells or tiny storage units where data is stored as bits, o's and 1's. And each cell consists of two main components, capacitors and transistors.

Capacitor: It is an electronical component that temporarily stores electrical energy in an electric field. Like a bucket that holds water, if the bucket is full, it represents a 1 and if it is empty, it represents a 0.

Transistor: It is a semiconductor device that acts as an electronic switch, managing the flow of a current. It acts as a gatekeeper that checks if the bucket is full or empty.

There are 2 types of RAM:

1.DRAM (Dynamic RAM**)**. The "Dynamic" part is a bit of a flaw: because these microscopic buckets have tiny leaks. To keep the data from disappearing, the computer must "refill" the full buckets thousands of times every second.

This is why when you turn off a computer, everything in the RAM is lost. The power stops, the refilling stops, and the buckets all leak dry.

2.SRAM (Static RAM)

It is used in very fast memory like CPU cache. Each bit uses: 6 transistors and it does NOT need refreshing.

An Analogy to understand it better:

Instead of a leaky bucket, it's like a light switch that stays on or off. It is faster, but it is much more expensive and takes more space.

How RAM communicates with the CPU:

RAM connects to the CPU through a Memory bus which is like a highway. When the CPU needs data it sends an address to the RAM. The Ram finds the location of the address send and retrieves the data requested by the CPU and then sends data back immediately through the memory bus to the CPU, then the CPU performs the tasks required using the data that the RAM retrieved.

How RAM is manufactured:

1.Silicon Wafer: It starts with silicon, The silicon is melted and grown into a long cylindrical shape, which is then sliced into paper thin circular disks called wafers.

2.Photolithography (Printing with light): Because the capacitors and transistors are too small to be built by hand, we use light. The wafer is then coated with a light-sensitive chemical. A "stencil" of the circuit design is placed over it. Then Ultraviolet light shines through the stencil, ''baking'' the circuit pattern onto the wafer.

3.Etching and Doping: The wafer is then bathed in chemicals that eat away the unbaked parts, leaving behind the intricate" roads" and "structures" of the memory cells. We then add specific impurities (Doping) to the silicon to control how electricity flows through it.

4.Testing and Packaging: A single wafer might contain hundreds of individual RAM chips. They are tested for "defects". The good chips are cut out, encased in protective plastic, and placed onto the green rectangular sticks you see inside a computer.

Why AI needs RAM:

Artificial Intelligence models are essentially massive mathematical formulas with billions of variables. That is why AI needs massive datasets and extremely fast memory access

Training AI is like trying to memorize a 10,000-page book in one sitting.

AI needs RAM because it works with huge amounts of data and must process it very quickly. When an AI system is running, it constantly loads data, calculations, and instructions into RAM so the processor can access them instantly. If there isn't enough RAM, the system slows down because it must keep fetching data from slower storage. More RAM allows AI to handle bigger tasks, learn faster, and make decisions more efficiently, which is why modern AI systems depend heavily on large and fast RAM

This has led to the creation of HBM (High Bandwidth Memory)—which is essentially several RAM chips stacked on top of each other like a skyscraper to save space and increase speed. For an investor, the companies capable of stacking these chips reliably are currently at the top of the food chain.

Section 3:

Intel	Snap Dragon	Mac Silicon
Instruction Set Architecture		
Uses the x86-64, a complex Instruction Set Computing (CISC) architecture, which is highly powerful for desktop and server performance.	Uses ARM, a Reduced Instruction Set Computing (RISC) architecture. Efficient, designed for mobile.	Also uses ARM, a Reduced Instruction Set Computing (RISC) architecture, but it is more customized for macOS and iOS apps.
Memory Design		
Intel traditionally used a separate system RAM and dedicated GPU VRAM, though the newer versions of Intel chips have started integrating memory onto the chip.	Snapdragon also uses separate system RAM and dedicated GPU VRAM.	Apple Silicon uses Unified Memory Architecture, where the CPU and GPU can share one high-speed pool of memory.
System on a Chip (SoC) Integration		
Intel historically separated these into different chips on a motherboard, but they are moving toward more of an integrated "tiled" design.	Snapdragon chips are true SoCs, with an integrated CPU, GPU, Neural Engine, and more. These chips are power efficient and have higher speed performance.	Apple chips are also true SoCs, integrating the CPU, GPU, Neural Engine, and more. They are power efficient and have higher speed performance.
Power Efficiency		
Intel typically requires more power, starting at roughly 28W, to reach the same performance levels as Apple and SnapDragon.	Snapdragon's power usage is a bit more efficient for Windows laptops. They typically operate using around 23W.	Apple Silicon has the highest efficiency, sometimes consuming as little as 2–20W.
Multi-Core Scalability		
Intel usually wins in raw multi-core throughput for desktops, as it can push much higher power (TDP). Intel uses a hybrid of multiple small efficiency cores, and a few larger performance cores.	SnapDragon uses only high-performance cores. An example of a Snapdragon CPU is the X Elite, which has 12 high-performance cores and can sometimes surpass Apple when it comes to multi-threaded tasks.	Apple Silicon uses the same cores as Intel but they use a lot less cores which are larger and more efficient.
Single-Core Speed		
Intel pushes it's performance cores to a much higher clock speed,	The Snapdragon X Elite has its own unique "Oryon" cores which are made to try	Apple Silicon consistently leads in single-core performance with their new

but this often leads to overheating.	and match Apples high performance cores. This allows SnapDragon to beat Intel with single-core speed.	the M4 chip. Its architecture allows it to execute more instructions without needing an extremely high clock speed.
Thermal Management		
Intel chips usually require active cooling fans for sustained high performance as they produce a lot of heat.	Because of their ARM roots, Snapdragon chips run much cooler. This means they don't need any external cooling hardware such as fans.	Apple Silicon also ARM, which means their chips run much cooler, allowing for fanless designs like their MacBook Air.
Graphics (GPU)		
Intel used to have a rather weak CPU, but their new Arc Graphics card is a lot more efficient for integrated solutions.	Snapdragon's Adreno GPU is mainly optimized for mobile efficiency. It is designed to provide the most "visuals per watt."	Apple uses a Unified Memory Architecture(UMA). In a Mac the GPU and CPU are next to each other, so they share the same pool of high-speed memory which allows faster communication.
Connectivity		
Intel relies on separate Wi-Fi and Cellular modules.	Snapdragon chips often include 5G/LTE modems which are integrated in the SoC.	Apple Macs also relies on separate Wi-Fi and Cellular modules.
App Compatibility		
Intel chips have a native compatibility with standard Windows and Linux software.	Snapdragon chips can run Windows on ARM.	Apple moved from Intel to ARM in 2020 and used a translation tool called Rosetta 2.
AI Acceleration (NPU)		
Intel is catching up with its latest AI PC initiatives.	The Snapdragon X Elite features a Hexagon NPU capable of 45 trillion of operations per second.	The newest Apple M4 chip features a 16-core Neural Engine that reaches 38 TOPS.
Target Ecosystem		
Intel is used in the Windows and Linux PC market.	Snapdragon is the dominant force in Android and is the primary driver for the new Copilot+ PC category for Windows.	Apple Silicon is strictly compatible to macOS and iPadOS.

Section 4: Storage

Introduction

Since there is always a trade-off between speed, cost efficiency and storage, multiple different tiers of storage are used. This is called tiered storage. The different tiers can be seen as the “Temperature” of data being used, the colder the data, less its being used, the hotter the data, the more its being used.

LTO-10 (Linear Tape-Open) – Magnetic Tape drives [Cold Storage]

As opposed to old tape drives that were common in the 1950’s – 1980’s that had little storage, the newest LTO technology allows for an average of 30 Terabyte (enough space to store between 1.4 million and 9 million+ photos). This makes LTO-10 the perfect way to store a large amount of data in a small space. a Long ribbon is used to store data; the ribbon is coated in a microscopic layer of metallic particles that are orientated to represent 0’s and 1’s. The only drawback is that the long ribbon needs to be read from start to finish every single time data needs to be read or written, in contrast to SSD’s & HDD’s that can access any data at any point. This takes long and it’s also the reason we switched to newer technology in the past, but modern LTO-9/10 moves the ribbon forward with speeds that exceeds 5m per second allowing us to read it faster than any old tape driver.

Recently Tape drivers made a comeback to the market due to the recent surge in ransomware (programs people write to hold all your companies records and sensitive documents for ransom). Ransomware became better with time, and this eventually led to it waiting long enough to infect all the companies’ backups and the extra backups preventing companies from simply restoring everything. The current best way to deal with ransomware is to simply create an “air gap” by removing the drive from the system and placing it in storage. Magnetic tapes can sit on a shelf for 10, 20 or even 30 years and it will still not lose any data.

Magnetic tapes also don’t need any source of power when they are idling in storage, in comparison to HDD that often is kept on constant power to avoid any mechanical stress from starting it up constantly. Magnetic Tapes also are more cost efficient than enterprise HDD especially in multi petabyte businesses (petabytes = 1000 Terabyte). The magnetic tape driver has a higher initial cost than a single HDD, but the cartridges are cheaper than HDD’s, significantly lowering operational costs. LTO reportedly reduced the cost of storage and floor space by 50-70%. In summary Magnetic Tape is slower but safer and it acts as a disaster recovery strategy against ransomware attacks. At the end of the day, magnetic tapes are like a box of sweets you store in a very high place away from children, it takes some effort to get it down again, but it is ultimately at the safest place possible.

Enterprise HDD (Hard Disk Drive) [Warm Storage]

HDD's use electromagnetism to manipulate the magnetic polarity of microscopic regions on the disk to change the direction it points, either up (1's) or down (0's). A current flows through a coil which inducts a magnetic field in the core of the write head. This core has a small break at the tip that forces the magnetic flux to jump across the gap causing something called a "fringing field" to leak out. When the actuator arm moves to a section while the platter beneath the head spins, the head will cause a fringing field to leak out and forces a specific spot to change direction. For the reading, a HDD uses a read head that is positioned with the actuator arm on the specific track, when it reaches a bit (specific spot on disk) with the same magnetic polarity, the resistance drops causing max current (1's), but when it reaches a bit with the opposite polarity the resistance spikes causing no current (0's) . a low voltage current is sent through the read head, and changes in the current are read as 0's and 1's.

HDDs are used to store any data that is accessed recently or that is being planned to be used. The main reason for this is because HDDs can access their information faster than LTO since it can just move the actuator arm to a point to read the information instead of reading the whole disk, and HDDs are cheaper than SSDs which makes it the perfect middle ground. HDD's platter spins at high speed, the higher the speed the lower the latency (delay), but higher speed comes with more heat and a higher mechanical stress.

HDD's usually last around 3-5 years. It can last up till 10 years, but the risk of mechanical failure increases significantly after 3 years due to the moving parts causing physical wear. To prevent mechanical failure HDDs are often kept on constant power to reduce the number of times the disk needs to start and end spinning. HDDs usually have mechanical stress on the bearings or on the motor during the acceleration phase. Power surges can also have an impact on its life span and if the disk is still spinning while the power goes down there is a risk for data corruption

A way to see an HDD is like an automated filing cabinet. A physical arm must first go to a specific file before accessing it to change it or read it. The same way a HDD must move its arm to a section to read a specific point. HDDs are the standard storage used as any external storage device for laptops or PCs due to its relatively low cost in comparison to SSDs and LTOs (accounting for the LTO driver which significantly raises cost if you are anyway using a single cartridge)

NVMe SSD (Solid State Drive) [Hot Storage]

NVMe stands for non-volatile memory express, and it is a protocol that allows the SSD to directly communicate with the CPU (central processing unit) using a faster route than normal, this protocol is essential to overcome the speed bottleneck that occurs normally with SSDs. SSDs have a unique way of storing data using transistors and filling them with electrons creating a NAND Flash Cell (the system resembles the logic gate circuit Not-And and is also named NAND). Each SSD drive has a System-on-a-Chip (SoC) that contains a processor that controls where an electron pulse will be sent, considering too many electron pulses on a single NAND flash cell might break it, the SoC keeps a record of which is used and how much they are used (the Wear level) and which cells aren't usable anymore (Bad Block). a Bit line is used to send an electron charge to a specific NAND flash cell, and the electron gets stuck inside the NAND flash cell. The NAND cells are lined up and due to this when reading the data, a high voltage is sent through all the cells except the one being measured (pass voltage) to force them to open, while a low voltage (Read Voltage) is applied to the measured cell. In newer system each cell can hold up till three levels of charges (3 bits), where the read voltage is replaced with several voltages to identify the tipping point at which the cell conducts a current resulting in a three-digit binary number (ex 101 / 110).

SSDs are used for their incredible speed and can reach up till 3500MB per second. This faster speed enhances user experience, for example in the gaming industry, cutscenes were cut out since the system didn't need to wait to load anymore since all assets are loaded fast enough. SSDs are normally used for Hot data that is constantly worked with. Since SSDs can directly communicate with the CPU it is most beneficially used for things that are needed constantly

SSD's have the highest cost per terabyte out of all three the storage methods discussed. The recent spike in AI growth also increased the price further due to high demand of LLM (Large learning modules) used in the training of AI. The Average Lifespan of an SSD is around 5-10 years although this depends on how much data was written on the SSD rather than its age. SSDs do not use any power when not in use but in most cases especially in industry they will always be in use, and they will use more power than a standard enterprise HDD.

An SSD is like a fast car, its not needed always but it's nice to have and it allows you to reach speeds not possible with other options. SSDs outperform every other storage method but due to their high cost it's recommended to use it only for the most important data instead of everything.

Technical comparison

Feature	LTO-10	HDD	SSD
Native Capacity	30-40 TB	36-44 TB	1-2 TB
Lifespan	10-20 Years	3-10 Years	5-10 Years
Cyber Defence	Air Gap	Software Orientated	Software Orientated
Search Speed	Seconds to minutes	Milliseconds	Microseconds

Link to Video Presentation

https://drive.google.com/file/d/1mXZ2Fw0NWkwocXcRZBeWfaFoPS8L12W0/view?usp=sharing